Exercise: Classification with the k-Nearest Neighbors Algorithm

Statement

We consider the following 5 instances, each described by 5 numerical attributes and a class:

Instance	A1	A2	A3	A4	A5	**Class**
X_1	3	5	4	6	1	1
X_2	6	10	3	2	2	2
X_3	8	3	4	2	6	3
X_4	2	1	4	3	6	3
X_5	5	5	1	4	8	2

We want to classify the instance:

$$X_6 = \langle 3, 12, 4, 7, 8 \rangle$$

a) Determine the class of X_6 using the K-NN algorithm with $k = 1$.

b) Determine the class of X_6 using the K-NN algorithm with $k = 3$.

Solution

Method

We use the Euclidean distance:

$$d(X, Y) = \sqrt{\sum_{i=1}^{5} (x_i - y_i)^2}$$

Distances

- $d(X_6, X_1) = \sqrt{0 + 49 + 0 + 1 + 49} = \sqrt{99} \approx 9.95$
- $d(X_6, X_2) = \sqrt{9 + 4 + 1 + 25 + 36} = \sqrt{75} \approx 8.66$
- $d(X_6, X_3) = \sqrt{25 + 81 + 0 + 25 + 4} = \sqrt{135} \approx 11.62$

- $d(X_6, X_4) = \sqrt{1 + 121 + 0 + 16 + 4} = \sqrt{142} \approx 11.91$
- $d(X_6, X_5) = \sqrt{4 + 49 + 9 + 9 + 0} = \sqrt{71} \approx 8.43$

a) For $k = 1$

Nearest neighbor: X_5 (distance 8.43), class 2.
Result: Class 2.

b) For $k = 3$

Nearest neighbors:

- X_5 (class 2)
- X_2 (class 2)
- X_1 (class 1)

Majority class: **2**.
Result: Class 2.